What Drives Poor Mental Health Across U.S. Communities? A County-Level Analysis of Social and Economic Determinants in the United States

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Disclaimer: Generative AI was used to assist with drafting report information and templating and writing code in this assignment; however, this writing was checked manually and edited by hand to ensure accuracy.

Source: OpenAI. (2026). *ChatGPT* [Large language model]. https://chatgpt.com/.

Data Processing (TODO: move to Quarto website)

Load and inspect PLACES mental health data

I begin by loading and inspecting the PLACES data so I can confirm its structure, identify the county-level mental health variables available, and isolate the outcome used in the analysis.

```
places <- read.csv("places_county_2025.csv")
```

```
dim(places)
```

```
[1] 3143  167
```

```
head(places)
```

```
  StateAbbr    StateDesc CountyName CountyFIPS TotalPopulation TotalPop18plus
1        KY     Kentucky      Trigg      21221          14,369         11,280
2        KY     Kentucky      Wayne      21231          19,580         15,608
3        KY     Kentucky Washington      21229          12,267          9,420
4        PA Pennsylvania      Perry      42099          46,083         36,495
5        PA Pennsylvania    Venango      42121          49,431         39,983
6        OK     Oklahoma    Haskell      40061          11,832          9,053
  ACCESS2_CrudePrev ACCESS2_Crude95CI ACCESS2_AdjPrev ACCESS2_Adj95CI
1                NA                                NA
2                NA                                NA
3                NA                                NA
4                NA                                NA
5                NA                                NA
6              11.6      ( 9.3, 14.1)            12.1    ( 9.7, 14.8)
  ARTHRITIS_CrudePrev ARTHRITIS_Crude95CI ARTHRITIS_AdjPrev ARTHRITIS_Adj95CI
1                  NA                                    NA
```

```
2                  NA                                    NA
3                  NA                                    NA
4                  NA                                    NA
5                  NA                                    NA
6                32.6        (29.0, 36.4)             26.9      (23.7, 30.2)
  BINGE_CrudePrev BINGE_Crude95CI BINGE_AdjPrev BINGE_Adj95CI BPHIGH_CrudePrev
1              NA                            NA                             NA
2              NA                            NA                             NA
3              NA                            NA                             NA
4              NA                            NA                             NA
5              NA                            NA                             NA
6            12.8    (10.3, 15.6)          14.7  (11.9, 17.9)             44.1
  BPHIGH_Crude95CI BPHIGH_AdjPrev BPHIGH_Adj95CI BPMED_CrudePrev
1                               NA                             NA
2                               NA                             NA
3                               NA                             NA
4                               NA                             NA
5                               NA                             NA
6     (39.7, 48.6)           38.2   (34.0, 42.6)            78.8
  BPMED_Crude95CI BPMED_AdjPrev BPMED_Adj95CI CANCER_CrudePrev CANCER_Crude95CI
1                            NA                             NA
2                            NA                             NA
3                            NA                             NA
4                            NA                             NA
5                            NA                             NA
6    (76.0, 81.3)          59.4  (54.9, 64.0)              9.2     ( 8.2, 10.2)
  CANCER_AdjPrev CANCER_Adj95CI CASTHMA_CrudePrev CASTHMA_Crude95CI
1             NA                                NA
2             NA                                NA
3             NA                                NA
4             NA                                NA
5             NA                                NA
6              7   ( 6.3,  7.8)              12.2      (10.8, 13.8)
  CASTHMA_AdjPrev CASTHMA_Adj95CI CHD_CrudePrev CHD_Crude95CI CHD_AdjPrev
1              NA                            NA                        NA
2              NA                            NA                        NA
3              NA                            NA                        NA
4              NA                            NA                        NA
5              NA                            NA                        NA
6            12.4    (10.9, 14.0)           9.5  ( 8.4, 10.6)         7.2
   CHD_Adj95CI CHECKUP_CrudePrev CHECKUP_Crude95CI CHECKUP_AdjPrev
1                            NA                                NA
2                            NA                                NA
3                            NA                                NA
4                            NA                                NA
5                            NA                                NA
6 ( 6.4,  8.1)              79.2      (75.7, 82.7)            76.3
  CHECKUP_Adj95CI CHOLSCREEN_CrudePrev CHOLSCREEN_Crude95CI CHOLSCREEN_AdjPrev
1                                   NA                                      NA
2                                   NA                                      NA
3                                   NA                                      NA
```

```
4                                   NA                                        NA
5                                   NA                                        NA
6    (72.3, 80.2)                  84.2          (82.1, 86.4)               81.2
  CHOLSCREEN_Adj95CI COLON_SCREEN_CrudePrev COLON_SCREEN_Crude95CI
1                                     63.7           (58.0, 69.0)
2                                     59.7           (53.6, 65.3)
3                                     66.9           (61.6, 71.8)
4                                     70.6           (65.8, 75.2)
5                                     67.3           (61.9, 72.2)
6       (78.5, 83.8)                  57.9           (52.2, 63.4)
  COLON_SCREEN_AdjPrev COLON_SCREEN_Adj95CI COPD_CrudePrev COPD_Crude95CI
1                 56.5         (50.8, 62.0)             NA
2                 52.9         (46.9, 58.5)             NA
3                 61.1         (55.7, 66.3)             NA
4                 63.8         (58.7, 68.6)             NA
5                 59.0         (53.5, 64.2)             NA
6                 52.0         (46.4, 57.6)           10.8   ( 9.3, 12.5)
  COPD_AdjPrev COPD_Adj95CI CSMOKING_CrudePrev CSMOKING_Crude95CI
1           NA                              NA
2           NA                              NA
3           NA                              NA
4           NA                              NA
5           NA                              NA
6          8.9 ( 7.7, 10.3)               20.7       (17.4, 24.1)
  CSMOKING_AdjPrev CSMOKING_Adj95CI DENTAL_CrudePrev DENTAL_Crude95CI
1               NA                              55.8     (52.0, 59.8)
2               NA                              47.5     (43.2, 51.8)
3               NA                              59.5     (55.5, 63.5)
4               NA                              63.7     (59.8, 67.3)
5               NA                              62.4     (58.4, 66.3)
6             21.3     (17.9, 24.8)             49.3     (45.3, 53.1)
  DENTAL_AdjPrev DENTAL_Adj95CI DEPRESSION_CrudePrev DEPRESSION_Crude95CI
1           56.3   (52.5, 60.3)                  NA
2           46.7   (42.4, 51.1)                  NA
3           59.4   (55.3, 63.3)                  NA
4           62.1   (58.1, 65.9)                  NA
5           61.8   (57.8, 65.7)                  NA
6           47.1   (43.2, 51.0)                 25.4         (21.7, 29.5)
  DEPRESSION_AdjPrev DEPRESSION_Adj95CI DIABETES_CrudePrev DIABETES_Crude95CI
1                 NA                                    NA
2                 NA                                    NA
3                 NA                                    NA
4                 NA                                    NA
5                 NA                                    NA
6               26.8       (22.9, 30.9)               15.4       (13.2, 17.6)
  DIABETES_AdjPrev DIABETES_Adj95CI GHLTH_CrudePrev GHLTH_Crude95CI
1               NA                                NA
2               NA                                NA
3               NA                                NA
4               NA                                NA
5               NA                                NA
```

```
6             12.4     (10.7, 14.2)            25.7    (22.1, 29.5)
  GHLTH_AdjPrev GHLTH_Adj95CI HIGHCHOL_CrudePrev HIGHCHOL_Crude95CI
1            NA                              NA
2            NA                              NA
3            NA                              NA
4            NA                              NA
5            NA                              NA
6          24.1  (20.5, 27.7)               41.4       (36.9, 46.0)
  HIGHCHOL_AdjPrev HIGHCHOL_Adj95CI LPA_CrudePrev LPA_Crude95CI LPA_AdjPrev
1               NA                              NA                        NA
2               NA                              NA                        NA
3               NA                              NA                        NA
4               NA                              NA                        NA
5               NA                              NA                        NA
6             34.3     (30.1, 38.9)          36.3  (30.4, 41.9)        34.4
   LPA_Adj95CI MAMMOUSE_CrudePrev MAMMOUSE_Crude95CI MAMMOUSE_AdjPrev
1                           69.5       (60.6, 77.7)             68.1
2                           68.2       (58.7, 76.9)             68.3
3                           76.0       (68.1, 82.7)             75.9
4                           77.0       (69.3, 83.5)             77.3
5                           77.8       (69.8, 84.2)             77.7
6 (28.8, 39.8)              64.1       (54.3, 73.2)             63.4
  MAMMOUSE_Adj95CI MHLTH_CrudePrev MHLTH_Crude95CI MHLTH_AdjPrev MHLTH_Adj95CI
1     (59.0, 76.6)              NA                           NA
2     (58.7, 77.1)              NA                           NA
3     (68.0, 82.7)              NA                           NA
4     (69.6, 83.8)              NA                           NA
5     (69.8, 84.2)              NA                           NA
6     (53.5, 72.7)            18.8    (16.6, 21.2)          20.6  (18.1, 23.1)
  OBESITY_CrudePrev OBESITY_Crude95CI OBESITY_AdjPrev OBESITY_Adj95CI
1                NA                                NA
2                NA                                NA
3                NA                                NA
4                NA                                NA
5                NA                                NA
6              42.2      (33.8, 51.0)            42.3    (33.9, 51.0)
  PHLTH_CrudePrev PHLTH_Crude95CI PHLTH_AdjPrev PHLTH_Adj95CI SLEEP_CrudePrev
1              NA                            NA                          40.0
2              NA                            NA                          41.1
3              NA                            NA                          38.1
4              NA                            NA                          36.9
5              NA                            NA                          37.3
6            17.2    (15.0, 19.6)          15.9  (13.7, 18.0)            40.7
  SLEEP_Crude95CI SLEEP_AdjPrev SLEEP_Adj95CI STROKE_CrudePrev STROKE_Crude95CI
1    (32.8, 47.2)          41.4  (34.2, 48.7)               NA
2    (33.6, 48.5)          42.4  (34.8, 49.9)               NA
3    (31.4, 45.3)          39.1  (32.4, 46.5)               NA
4    (30.1, 44.2)          38.1  (31.2, 45.5)               NA
5    (30.3, 44.5)          38.8  (31.7, 46.2)               NA
6    (33.6, 48.1)          42.2  (35.0, 49.8)              4.9     ( 4.3,  5.4)
  STROKE_AdjPrev STROKE_Adj95CI TEETHLOST_CrudePrev TEETHLOST_Crude95CI
```

```
1             NA                                23.9        (20.1, 28.1)
2             NA                                29.5        (24.4, 35.2)
3             NA                                20.9        (17.2, 25.0)
4             NA                                14.3        (11.5, 17.4)
5             NA                                11.8        ( 9.4, 14.4)
6            3.8   ( 3.4,  4.2)                26.6        (22.5, 31.1)
  TEETHLOST_AdjPrev TEETHLOST_Adj95CI HEARING_CrudePrev HEARING_Crude95CI
1              23.4      (19.6, 27.5)                NA
2              29.8      (24.7, 35.4)                NA
3              20.9      (17.2, 25.0)                NA
4              14.7      (11.8, 17.9)                NA
5              12.1      ( 9.7, 14.7)                NA
6              26.7      (22.7, 31.2)              11.9      (10.5, 13.3)
  HEARING_AdjPrev HEARING_Adj95CI VISION_CrudePrev VISION_Crude95CI
1              NA                                NA
2              NA                                NA
3              NA                                NA
4              NA                                NA
5              NA                                NA
6             9.6    ( 8.5, 10.7)              7.6     ( 6.7,  8.5)
  VISION_AdjPrev VISION_Adj95CI COGNITION_CrudePrev COGNITION_Crude95CI
1             NA                                 NA
2             NA                                 NA
3             NA                                 NA
4             NA                                 NA
5             NA                                 NA
6            6.9   ( 6.1,  7.8)                19.2        (16.8, 21.8)
  COGNITION_AdjPrev COGNITION_Adj95CI MOBILITY_CrudePrev MOBILITY_Crude95CI
1                NA                                   NA
2                NA                                   NA
3                NA                                   NA
4                NA                                   NA
5                NA                                   NA
6              20.4      (17.7, 23.2)               21.7       (18.8, 24.8)
  MOBILITY_AdjPrev MOBILITY_Adj95CI SELFCARE_CrudePrev SELFCARE_Crude95CI
1               NA                                 NA
2               NA                                 NA
3               NA                                 NA
4               NA                                 NA
5               NA                                 NA
6             17.9     (15.4, 20.6)                5.4       ( 4.7,  6.2)
  SELFCARE_AdjPrev SELFCARE_Adj95CI INDEPLIVE_CrudePrev INDEPLIVE_Crude95CI
1               NA                                  NA
2               NA                                  NA
3               NA                                  NA
4               NA                                  NA
5               NA                                  NA
6              4.8     ( 4.2,  5.4)                11.6        (10.1, 13.3)
  INDEPLIVE_AdjPrev INDEPLIVE_Adj95CI DISABILITY_CrudePrev DISABILITY_Crude95CI
1                NA                                     NA
2                NA                                     NA
```

```
3                NA                                      NA
4                NA                                      NA
5                NA                                      NA
6              11.3      ( 9.8, 12.9)                    44         (38.6, 49.2)
  DISABILITY_AdjPrev DISABILITY_Adj95CI LONELINESS_CrudePrev
1                 NA                                      NA
2                 NA                                      NA
3                 NA                                      NA
4                 NA                                      NA
5                 NA                                      NA
6               41.5       (36.3, 46.8)                 32.3
  LONELINESS_Crude95CI LONELINESS_AdjPrev LONELINESS_Adj95CI
1                                     NA
2                                     NA
3                                     NA
4                                     NA
5                                     NA
6         (27.1, 37.7)               33.9       (28.6, 39.4)
  FOODSTAMP_CrudePrev FOODSTAMP_Crude95ci FOODSTAMP_AdjPrev FOODSTAMP_Adj95CI
1                  NA                                    NA
2                  NA                                    NA
3                  NA                                    NA
4                  NA                                    NA
5                  NA                                    NA
6                18.8        (15.7, 22.4)                 20      (16.7, 23.7)
  FOODINSECU_CrudePrev FOODINSECU_Crude95CI FOODINSECU_AdjPrev
1                   NA                                      NA
2                   NA                                      NA
3                   NA                                      NA
4                   NA                                      NA
5                   NA                                      NA
6                 21.3         (18.0, 25.1)                22.7
  FOODINSECU_Adj95CI HOUSINSECU_CrudePrev HOUSINSECU_Crude95CI
1                                     NA
2                                     NA
3                                     NA
4                                     NA
5                                     NA
6       (19.1, 26.6)                 14.8         (12.6, 17.3)
  HOUSINSECU_AdjPrev HOUSINSECU_Adj95CI SHUTUTILITY_CrudePrev
1                 NA                                       NA
2                 NA                                       NA
3                 NA                                       NA
4                 NA                                       NA
5                 NA                                       NA
6               16.7       (14.1, 19.4)                   11.4
  SHUTUTILITY_Crude95CI SHUTUTILITY_AdjPrev SHUTUTILITY_Adj95CI
1                                       NA
2                                       NA
3                                       NA
4                                       NA
```

```
5                                        NA
6          ( 9.8, 13.2)                12.8        (11.0, 14.7)
  LACKTRPT_CrudePrev LACKTRPT_Crude95CI LACKTRPT_AdjPrev LACKTRPT_Adj95CI
1                 NA                                  NA
2                 NA                                  NA
3                 NA                                  NA
4                 NA                                  NA
5                 NA                                  NA
6               10.9       ( 9.5, 12.5)             11.9     (10.3, 13.7)
  EMOTIONSPT_CrudePrev EMOTIONSPT_Crude95CI EMOTIONSPT_AdjPrev
1                   NA                                     NA
2                   NA                                     NA
3                   NA                                     NA
4                   NA                                     NA
5                   NA                                     NA
6                 25.9         (22.1, 30.1)                 27
  EMOTIONSPT_Adj95CI                                Geolocation
1                    POINT (-87.8732998555619 36.8063370344073)
2                    POINT (-84.8283531236799 36.8008845409014)
3                     POINT (-85.175051728938 37.7532982899311)
4                    POINT (-77.2622712474235 40.3984083146619)
5                    POINT (-79.7581868804174 41.4006288227586)
6       (22.9, 31.3)   POINT (-95.116494065308 35.224815744206)
```

```r
colnames(places)
```

```
 [1] "StateAbbr"              "StateDesc"              "CountyName"
 [4] "CountyFIPS"             "TotalPopulation"        "TotalPop18plus"
 [7] "ACCESS2_CrudePrev"      "ACCESS2_Crude95CI"      "ACCESS2_AdjPrev"
[10] "ACCESS2_Adj95CI"        "ARTHRITIS_CrudePrev"    "ARTHRITIS_Crude95CI"
[13] "ARTHRITIS_AdjPrev"      "ARTHRITIS_Adj95CI"      "BINGE_CrudePrev"
[16] "BINGE_Crude95CI"        "BINGE_AdjPrev"          "BINGE_Adj95CI"
[19] "BPHIGH_CrudePrev"       "BPHIGH_Crude95CI"       "BPHIGH_AdjPrev"
[22] "BPHIGH_Adj95CI"         "BPMED_CrudePrev"        "BPMED_Crude95CI"
[25] "BPMED_AdjPrev"          "BPMED_Adj95CI"          "CANCER_CrudePrev"
[28] "CANCER_Crude95CI"       "CANCER_AdjPrev"         "CANCER_Adj95CI"
[31] "CASTHMA_CrudePrev"      "CASTHMA_Crude95CI"      "CASTHMA_AdjPrev"
[34] "CASTHMA_Adj95CI"        "CHD_CrudePrev"          "CHD_Crude95CI"
[37] "CHD_AdjPrev"            "CHD_Adj95CI"            "CHECKUP_CrudePrev"
[40] "CHECKUP_Crude95CI"      "CHECKUP_AdjPrev"        "CHECKUP_Adj95CI"
[43] "CHOLSCREEN_CrudePrev"   "CHOLSCREEN_Crude95CI"   "CHOLSCREEN_AdjPrev"
[46] "CHOLSCREEN_Adj95CI"     "COLON_SCREEN_CrudePrev" "COLON_SCREEN_Crude95CI"
[49] "COLON_SCREEN_AdjPrev"   "COLON_SCREEN_Adj95CI"   "COPD_CrudePrev"
[52] "COPD_Crude95CI"         "COPD_AdjPrev"           "COPD_Adj95CI"
[55] "CSMOKING_CrudePrev"     "CSMOKING_Crude95CI"     "CSMOKING_AdjPrev"
[58] "CSMOKING_Adj95CI"       "DENTAL_CrudePrev"       "DENTAL_Crude95CI"
[61] "DENTAL_AdjPrev"         "DENTAL_Adj95CI"         "DEPRESSION_CrudePrev"
[64] "DEPRESSION_Crude95CI"   "DEPRESSION_AdjPrev"     "DEPRESSION_Adj95CI"
[67] "DIABETES_CrudePrev"     "DIABETES_Crude95CI"     "DIABETES_AdjPrev"
[70] "DIABETES_Adj95CI"       "GHLTH_CrudePrev"        "GHLTH_Crude95CI"
```

```
 [73] "GHLTH_AdjPrev"          "GHLTH_Adj95CI"          "HIGHCHOL_CrudePrev"
 [76] "HIGHCHOL_Crude95CI"     "HIGHCHOL_AdjPrev"       "HIGHCHOL_Adj95CI"
 [79] "LPA_CrudePrev"          "LPA_Crude95CI"          "LPA_AdjPrev"
 [82] "LPA_Adj95CI"            "MAMMOUSE_CrudePrev"     "MAMMOUSE_Crude95CI"
 [85] "MAMMOUSE_AdjPrev"       "MAMMOUSE_Adj95CI"       "MHLTH_CrudePrev"
 [88] "MHLTH_Crude95CI"        "MHLTH_AdjPrev"          "MHLTH_Adj95CI"
 [91] "OBESITY_CrudePrev"      "OBESITY_Crude95CI"      "OBESITY_AdjPrev"
 [94] "OBESITY_Adj95CI"        "PHLTH_CrudePrev"        "PHLTH_Crude95CI"
 [97] "PHLTH_AdjPrev"          "PHLTH_Adj95CI"          "SLEEP_CrudePrev"
[100] "SLEEP_Crude95CI"        "SLEEP_AdjPrev"          "SLEEP_Adj95CI"
[103] "STROKE_CrudePrev"       "STROKE_Crude95CI"       "STROKE_AdjPrev"
[106] "STROKE_Adj95CI"         "TEETHLOST_CrudePrev"    "TEETHLOST_Crude95CI"
[109] "TEETHLOST_AdjPrev"      "TEETHLOST_Adj95CI"      "HEARING_CrudePrev"
[112] "HEARING_Crude95CI"      "HEARING_AdjPrev"        "HEARING_Adj95CI"
[115] "VISION_CrudePrev"       "VISION_Crude95CI"       "VISION_AdjPrev"
[118] "VISION_Adj95CI"         "COGNITION_CrudePrev"    "COGNITION_Crude95CI"
[121] "COGNITION_AdjPrev"      "COGNITION_Adj95CI"      "MOBILITY_CrudePrev"
[124] "MOBILITY_Crude95CI"     "MOBILITY_AdjPrev"       "MOBILITY_Adj95CI"
[127] "SELFCARE_CrudePrev"     "SELFCARE_Crude95CI"     "SELFCARE_AdjPrev"
[130] "SELFCARE_Adj95CI"       "INDEPLIVE_CrudePrev"    "INDEPLIVE_Crude95CI"
[133] "INDEPLIVE_AdjPrev"      "INDEPLIVE_Adj95CI"      "DISABILITY_CrudePrev"
[136] "DISABILITY_Crude95CI"   "DISABILITY_AdjPrev"     "DISABILITY_Adj95CI"
[139] "LONELINESS_CrudePrev"   "LONELINESS_Crude95CI"   "LONELINESS_AdjPrev"
[142] "LONELINESS_Adj95CI"     "FOODSTAMP_CrudePrev"    "FOODSTAMP_Crude95ci"
[145] "FOODSTAMP_AdjPrev"      "FOODSTAMP_Adj95CI"      "FOODINSECU_CrudePrev"
[148] "FOODINSECU_Crude95CI"   "FOODINSECU_AdjPrev"     "FOODINSECU_Adj95CI"
[151] "HOUSINSECU_CrudePrev"   "HOUSINSECU_Crude95CI"   "HOUSINSECU_AdjPrev"
[154] "HOUSINSECU_Adj95CI"     "SHUTUTILITY_CrudePrev"  "SHUTUTILITY_Crude95CI"
[157] "SHUTUTILITY_AdjPrev"    "SHUTUTILITY_Adj95CI"    "LACKTRPT_CrudePrev"
[160] "LACKTRPT_Crude95CI"     "LACKTRPT_AdjPrev"       "LACKTRPT_Adj95CI"
[163] "EMOTIONSPT_CrudePrev"   "EMOTIONSPT_Crude95CI"   "EMOTIONSPT_AdjPrev"
[166] "EMOTIONSPT_Adj95CI"     "Geolocation"
```

```r
str(places)
```

```
'data.frame':   3143 obs. of  167 variables:
 $ StateAbbr             : chr  "KY" "KY" "KY" "PA" ...
 $ StateDesc             : chr  "Kentucky" "Kentucky" "Kentucky" "Pennsylvania" ...
 $ CountyName            : chr  "Trigg" "Wayne" "Washington" "Perry" ...
 $ CountyFIPS            : int  21221 21231 21229 42099 42121 40061 17097 20005 2020 40133 ...
 $ TotalPopulation       : chr  "14,369" "19,580" "12,267" "46,083" ...
 $ TotalPop18plus        : chr  "11,280" "15,608" "9,420" "36,495" ...
 $ ACCESS2_CrudePrev     : num  NA NA NA NA NA 11.6 9.2 8.8 9.5 12.3 ...
 $ ACCESS2_Crude95CI     : chr  "" "" "" "" ...
 $ ACCESS2_AdjPrev       : num  NA NA NA NA NA 12.1 9.7 9.3 9.5 12.9 ...
 $ ACCESS2_Adj95CI       : chr  "" "" "" "" ...
 $ ARTHRITIS_CrudePrev   : num  NA NA NA NA NA 32.6 22 26.9 22.1 32.6 ...
 $ ARTHRITIS_Crude95CI   : chr  "" "" "" "" ...
 $ ARTHRITIS_AdjPrev     : num  NA NA NA NA NA 26.9 19.3 24.4 22 28.3 ...
 $ ARTHRITIS_Adj95CI     : chr  "" "" "" "" ...
```

```
 $ BINGE_CrudePrev       : num  NA NA NA NA NA 12.8 15.8 16.5 18 13 ...
 $ BINGE_Crude95CI       : chr  "" "" "" "" ...
 $ BINGE_AdjPrev         : num  NA NA NA NA NA 14.7 17 17.8 18 14.4 ...
 $ BINGE_Adj95CI         : chr  "" "" "" "" ...
 $ BPHIGH_CrudePrev      : num  NA NA NA NA NA 44.1 32.4 34.8 31.5 44.4 ...
 $ BPHIGH_Crude95CI      : chr  "" "" "" "" ...
 $ BPHIGH_AdjPrev        : num  NA NA NA NA NA 38.2 29.4 32.6 31.4 40.1 ...
 $ BPHIGH_Adj95CI        : chr  "" "" "" "" ...
 $ BPMED_CrudePrev       : num  NA NA NA NA NA 78.8 74.8 78.3 67.6 77.8 ...
 $ BPMED_Crude95CI       : chr  "" "" "" "" ...
 $ BPMED_AdjPrev         : num  NA NA NA NA NA 59.4 55.4 59.8 51.6 60.1 ...
 $ BPMED_Adj95CI         : chr  "" "" "" "" ...
 $ CANCER_CrudePrev      : num  NA NA NA NA NA 9.2 7.7 8.3 6.5 8.5 ...
 $ CANCER_Crude95CI      : chr  "" "" "" "" ...
 $ CANCER_AdjPrev        : num  NA NA NA NA NA 7 6.6 7 6.5 6.9 ...
 $ CANCER_Adj95CI        : chr  "" "" "" "" ...
 $ CASTHMA_CrudePrev     : num  NA NA NA NA NA 12.2 9.1 10.4 10.7 12.7 ...
 $ CASTHMA_Crude95CI     : chr  "" "" "" "" ...
 $ CASTHMA_AdjPrev       : num  NA NA NA NA NA 12.4 9.1 10.4 10.6 12.8 ...
 $ CASTHMA_Adj95CI       : chr  "" "" "" "" ...
 $ CHD_CrudePrev         : num  NA NA NA NA NA 9.5 5.4 6.9 5.1 9.6 ...
 $ CHD_Crude95CI         : chr  "" "" "" "" ...
 $ CHD_AdjPrev           : num  NA NA NA NA NA 7.2 4.6 5.8 5.1 7.8 ...
 $ CHD_Adj95CI           : chr  "" "" "" "" ...
 $ CHECKUP_CrudePrev     : num  NA NA NA NA NA 79.2 76.9 75.2 70.7 79.1 ...
 $ CHECKUP_Crude95CI     : chr  "" "" "" "" ...
 $ CHECKUP_AdjPrev       : num  NA NA NA NA NA 76.3 75.1 73.6 70.7 77 ...
 $ CHECKUP_Adj95CI       : chr  "" "" "" "" ...
 $ CHOLSCREEN_CrudePrev  : num  NA NA NA NA NA 84.2 88.2 81.6 80.4 82.8 ...
 $ CHOLSCREEN_Crude95CI  : chr  "" "" "" "" ...
 $ CHOLSCREEN_AdjPrev    : num  NA NA NA NA NA 81.2 87 81.4 80.6 80.7 ...
 $ CHOLSCREEN_Adj95CI    : chr  "" "" "" "" ...
 $ COLON_SCREEN_CrudePrev: num  63.7 59.7 66.9 70.6 67.3 57.9 61.9 61.6 60.4 55.1 ...
 $ COLON_SCREEN_Crude95CI: chr  "(58.0, 69.0)" "(53.6, 65.3)" "(61.6, 71.8)" "(65.8, 75.2)" ...
 $ COLON_SCREEN_AdjPrev  : num  56.5 52.9 61.1 63.8 59 52 57.1 56 55.9 49.2 ...
 $ COLON_SCREEN_Adj95CI  : chr  "(50.8, 62.0)" "(46.9, 58.5)" "(55.7, 66.3)" "(58.7, 68.6)" ...
 $ COPD_CrudePrev        : num  NA NA NA NA NA 10.8 5.4 7.6 5.3 11.4 ...
 $ COPD_Crude95CI        : chr  "" "" "" "" ...
 $ COPD_AdjPrev          : num  NA NA NA NA NA 8.9 4.7 6.8 5.2 9.9 ...
 $ COPD_Adj95CI          : chr  "" "" "" "" ...
 $ CSMOKING_CrudePrev    : num  NA NA NA NA NA 20.7 10.4 15.3 14 22.5 ...
 $ CSMOKING_Crude95CI    : chr  "" "" "" "" ...
 $ CSMOKING_AdjPrev      : num  NA NA NA NA NA 21.3 10.5 17 14.1 23.3 ...
 $ CSMOKING_Adj95CI      : chr  "" "" "" "" ...
 $ DENTAL_CrudePrev      : num  55.8 47.5 59.5 63.7 62.4 49.3 64.3 62.9 64.2 49.9 ...
 $ DENTAL_Crude95CI      : chr  "(52.0, 59.8)" "(43.2, 51.8)" "(55.5, 63.5)" "(59.8, 67.3)" ...
 $ DENTAL_AdjPrev        : num  56.3 46.7 59.4 62.1 61.8 47.1 63.6 60.7 64.2 48.2 ...
 $ DENTAL_Adj95CI        : chr  "(52.5, 60.3)" "(42.4, 51.1)" "(55.3, 63.3)" "(58.1, 65.9)" ...
 $ DEPRESSION_CrudePrev  : num  NA NA NA NA NA 25.4 18.1 21.7 21.3 27.6 ...
 $ DEPRESSION_Crude95CI  : chr  "" "" "" "" ...
 $ DEPRESSION_AdjPrev    : num  NA NA NA NA NA 26.8 18.5 22.3 21.2 28.7 ...
```

```
 $ DEPRESSION_Adj95CI    : chr  "" "" "" "" ...
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 $ DIABETES_Crude95CI    : chr  "" "" "" "" ...
 $ DIABETES_AdjPrev      : num  NA NA NA NA NA 12.4 9.8 10 8.8 13.7 ...
 $ DIABETES_Adj95CI      : chr  "" "" "" "" ...
 $ GHLTH_CrudePrev       : num  NA NA NA NA NA 25.7 16 19 17.5 27.6 ...
 $ GHLTH_Crude95CI       : chr  "" "" "" "" ...
 $ GHLTH_AdjPrev         : num  NA NA NA NA NA 24.1 15.4 18.6 17.5 26.4 ...
 $ GHLTH_Adj95CI         : chr  "" "" "" "" ...
 $ HIGHCHOL_CrudePrev    : num  NA NA NA NA NA 41.4 36.8 35.2 31.5 41.8 ...
 $ HIGHCHOL_Crude95CI    : chr  "" "" "" "" ...
 $ HIGHCHOL_AdjPrev      : num  NA NA NA NA NA 34.3 32.7 31 29.5 35.7 ...
 $ HIGHCHOL_Adj95CI      : chr  "" "" "" "" ...
 $ LPA_CrudePrev         : num  NA NA NA NA NA 36.3 19.9 27.7 19.8 37.9 ...
 $ LPA_Crude95CI         : chr  "" "" "" "" ...
 $ LPA_AdjPrev           : num  NA NA NA NA NA 34.4 19.4 27.4 20 36.7 ...
 $ LPA_Adj95CI           : chr  "" "" "" "" ...
 $ MAMMOUSE_CrudePrev    : num  69.5 68.2 76 77 77.8 64.1 75.5 69.3 69.7 65.5 ...
 $ MAMMOUSE_Crude95CI    : chr  "(60.6, 77.7)" "(58.7, 76.9)" "(68.1, 82.7)" "(69.3, 83.5)" ...
 $ MAMMOUSE_AdjPrev      : num  68.1 68.3 75.9 77.3 77.7 63.4 75.1 68.9 69 63.9 ...
 $ MAMMOUSE_Adj95CI      : chr  "(59.0, 76.6)" "(58.7, 77.1)" "(68.0, 82.7)" "(69.6, 83.8)" ...
 $ MHLTH_CrudePrev       : num  NA NA NA NA NA 18.8 14.1 17.8 16.1 20.3 ...
 $ MHLTH_Crude95CI       : chr  "" "" "" "" ...
 $ MHLTH_AdjPrev         : num  NA NA NA NA NA 20.6 14.7 18.5 16.1 21.7 ...
 $ MHLTH_Adj95CI         : chr  "" "" "" "" ...
 $ OBESITY_CrudePrev     : num  NA NA NA NA NA 42.2 32.9 37 35 45 ...
 $ OBESITY_Crude95CI     : chr  "" "" "" "" ...
 $ OBESITY_AdjPrev       : num  NA NA NA NA NA 42.3 32.9 38.8 35 45.4 ...
 $ OBESITY_Adj95CI       : chr  "" "" "" "" ...
 $ PHLTH_CrudePrev       : num  NA NA NA NA NA 17.2 11 13 12.1 18.2 ...
 $ PHLTH_Crude95CI       : chr  "" "" "" "" ...
 $ PHLTH_AdjPrev         : num  NA NA NA NA NA 15.9 10.4 12.6 12.1 17.2 ...
 $ PHLTH_Adj95CI         : chr  "" "" "" "" ...
 $ SLEEP_CrudePrev       : num  40 41.1 38.1 36.9 37.3 40.7 33.1 34.4 34.6 39.6 ...
  [list output truncated]
```

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
places_mh <- places %>%
  select(
    StateAbbr,
    StateDesc,
    CountyName,
    CountyFIPS,
    TotalPopulation,
    TotalPop18plus,
    MHLTH_AdjPrev,
    MHLTH_Adj95CI
  ) %>%
  rename(
    poor_mental_health = MHLTH_AdjPrev,
    poor_mental_health_ci = MHLTH_Adj95CI
  )
```

```r
dim(places_mh)
```

```
[1] 3143    8
```

```r
head(places_mh)
```

```
  StateAbbr    StateDesc CountyName CountyFIPS TotalPopulation TotalPop18plus
1        KY     Kentucky      Trigg      21221          14,369         11,280
2        KY     Kentucky      Wayne      21231          19,580         15,608
3        KY     Kentucky Washington      21229          12,267          9,420
4        PA Pennsylvania      Perry      42099          46,083         36,495
5        PA Pennsylvania    Venango      42121          49,431         39,983
6        OK     Oklahoma    Haskell      40061          11,832          9,053
  poor_mental_health poor_mental_health_ci
1                 NA
2                 NA
3                 NA
4                 NA
5                 NA
6               20.6          (18.1, 23.1)
```

```r
str(places_mh)
```

```
'data.frame':   3143 obs. of  8 variables:
 $ StateAbbr            : chr  "KY" "KY" "KY" "PA" ...
 $ StateDesc            : chr  "Kentucky" "Kentucky" "Kentucky" "Pennsylvania" ...
 $ CountyName           : chr  "Trigg" "Wayne" "Washington" "Perry" ...
 $ CountyFIPS           : int  21221 21231 21229 42099 42121 40061 17097 20005 2020 40133 ...
 $ TotalPopulation      : chr  "14,369" "19,580" "12,267" "46,083" ...
 $ TotalPop18plus       : chr  "11,280" "15,608" "9,420" "36,495" ...
 $ poor_mental_health   : num  NA NA NA NA NA 20.6 14.7 18.5 16.1 21.7 ...
 $ poor_mental_health_ci: chr  "" "" "" "" ...
```

Load and inspect ACS predictors

I next load the ACS data to obtain the county-level social and economic predictors and prepare them for merging with the PLACES mental health outcome.

```
library(tidycensus)

api_key <- Sys.getenv("CENSUS_API_KEY")
census_api_key(api_key, install = FALSE)
```

```
To install your API key for use in future sessions, run this function with `install = TRUE`.
```

```
library(tidycensus)
library(dplyr)
library(tidyr)

api_key <- Sys.getenv("CENSUS_API_KEY")
census_api_key(api_key, install = FALSE)
```

```
To install your API key for use in future sessions, run this function with `install = TRUE`.
```

```
acs_raw <- get_acs(
  geography = "county",
  year = 2023,
  survey = "acs5",
  variables = c(
    poverty_rate = "S1701_C03_001",
    median_household_income = "B19013_001",
    unemployment_rate = "DP03_0009PE",
    bachelors_or_higher = "DP02_0068PE",
    uninsured_rate = "S2701_C05_001",
    rent_burden_30_34_9 = "DP04_0141PE",
    rent_burden_35_plus = "DP04_0142PE"
  )
)
```

```
Getting data from the 2019-2023 5-year ACS

Fetching data by table type ("B/C", "S", "DP") and combining the result.
```

```
unique(acs_raw$variable)
```

```
[1] "median_household_income" "poverty_rate"
[3] "uninsured_rate"          "DP02_0068P"
[5] "DP03_0009P"              "DP04_0141P"
[7] "DP04_0142P"
```

```
acs_county <- acs_raw %>%
  select(GEOID, NAME, variable, estimate) %>%
  pivot_wider(names_from = variable, values_from = estimate) %>%
  rename(
    bachelors_or_higher = DP02_0068P,
    unemployment_rate = DP03_0009P,
    rent_burden_30_34_9 = DP04_0141P,
    rent_burden_35_plus = DP04_0142P
  ) %>%
  mutate(
    housing_cost_burden = rent_burden_30_34_9 + rent_burden_35_plus
  ) %>%
  select(
    GEOID,
    NAME,
    poverty_rate,
    median_household_income,
    unemployment_rate,
    bachelors_or_higher,
    uninsured_rate,
    rent_burden_30_34_9,
    rent_burden_35_plus,
    housing_cost_burden
  )
```

```
places_mh <- places_mh %>%
  mutate(CountyFIPS = as.character(CountyFIPS))

acs_county <- acs_county %>%
  mutate(GEOID = as.character(GEOID))
```

```
analysis_df <- places_mh %>%
  left_join(acs_county, by = c("CountyFIPS" = "GEOID"))
```

Load and inspect Census GEOIDs and TIGER/Line shapefiles

Finally, I load county geographic identifiers and shapefiles so the merged dataset can be mapped and explored spatially across U.S. counties.

```
library(tigris)
```

```
To enable caching of data, set `options(tigris_use_cache = TRUE)`
in your R script or .Rprofile.
```

```
library(dplyr)
```

```
counties_sf <- counties(year = 2024, cb = TRUE) %>%
  mutate(GEOID = as.character(GEOID))
```

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```

```
analysis_df <- analysis_df %>%
  mutate(CountyFIPS = as.character(CountyFIPS))

map_df <- counties_sf %>%
  left_join(analysis_df, by = c("GEOID" = "CountyFIPS"))
```

EDA plots

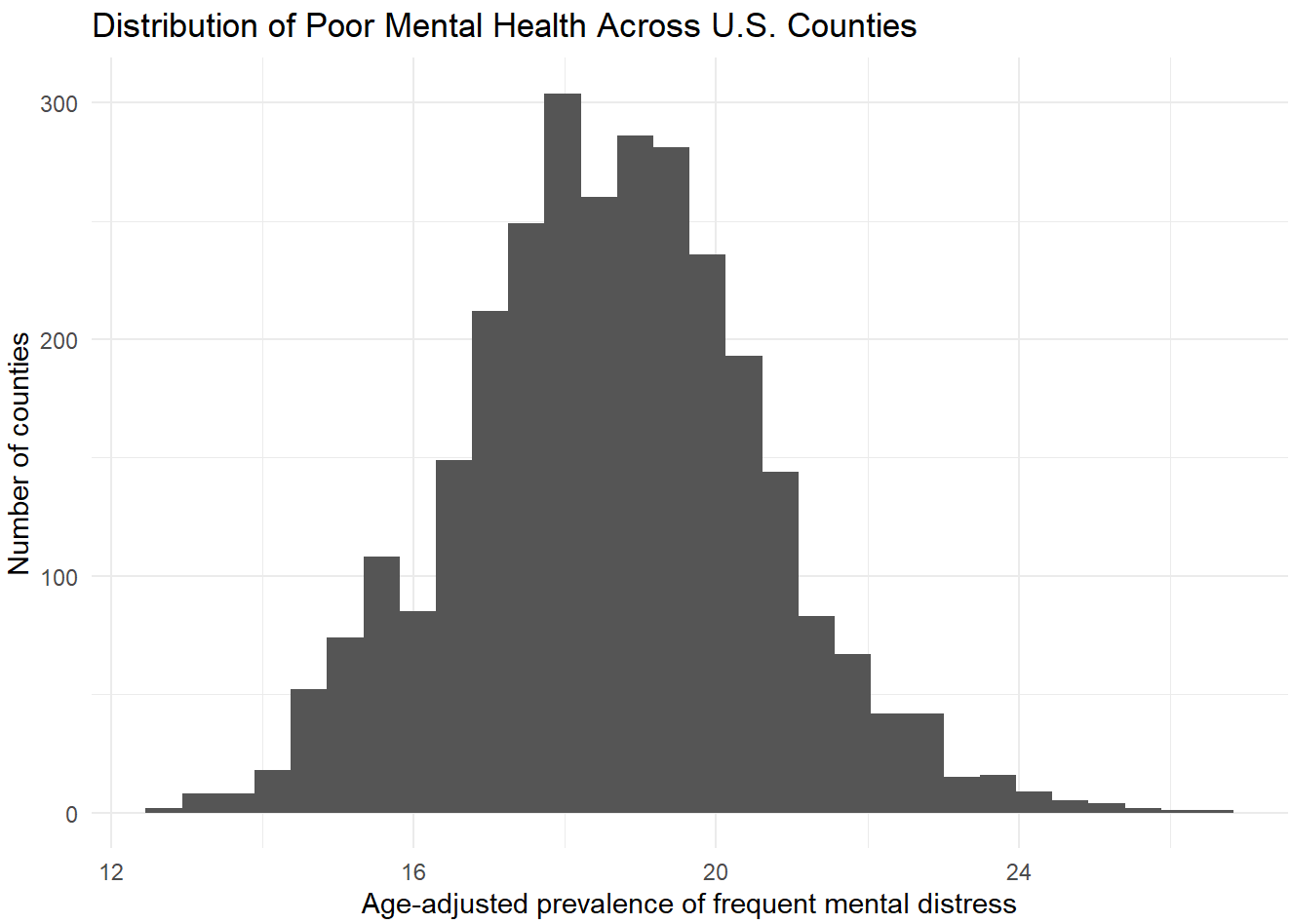
To begin exploration, I first examine the distribution of the outcome variable. This helps show how poor mental health is distributed across counties and whether there are signs of skewness, clustering, or unusual values.

```
# Outcome distribution
library(ggplot2)
```

```
library(dplyr)

ggplot(analysis_df, aes(x = poor_mental_health)) +
  geom_histogram(bins = 30) +
  labs(
    title = "Distribution of Poor Mental Health Across U.S. Counties",
    x = "Age-adjusted prevalence of frequent mental distress",
    y = "Number of counties"
  ) +
  theme_minimal()
```

```
Warning: Removed 187 rows containing non-finite outside the scale range
(`stat_bin()`).
```



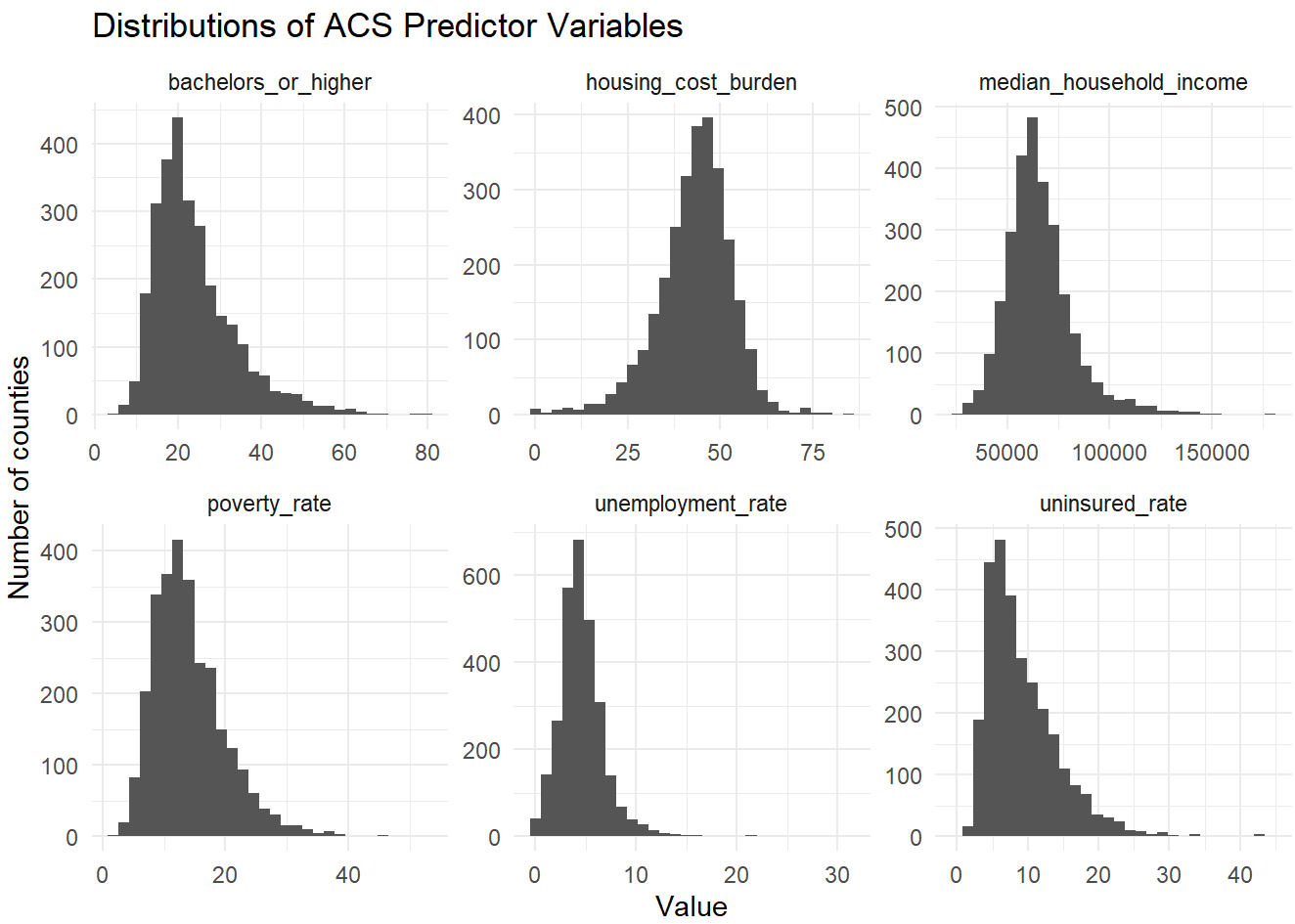
Next, I examine the distributions of the ACS predictor variables. This helps identify their ranges and variation across counties and can reveal whether any predictors are especially skewed or concentrated.

```
## Predictor distributions
library(tidyr)
library(ggplot2)
library(dplyr)

analysis_df %>%
  select(
```

```
    poverty_rate,
    median_household_income,
    unemployment_rate,
    bachelors_or_higher,
    uninsured_rate,
    housing_cost_burden
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = "variable",
    values_to = "value"
  ) %>%
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ variable, scales = "free") +
  labs(
    title = "Distributions of ACS Predictor Variables",
    x = "Value",
    y = "Number of counties"
  ) +
  theme_minimal()
```

```
Warning: Removed 1912 rows containing non-finite outside the scale range
(`stat_bin()`).
```



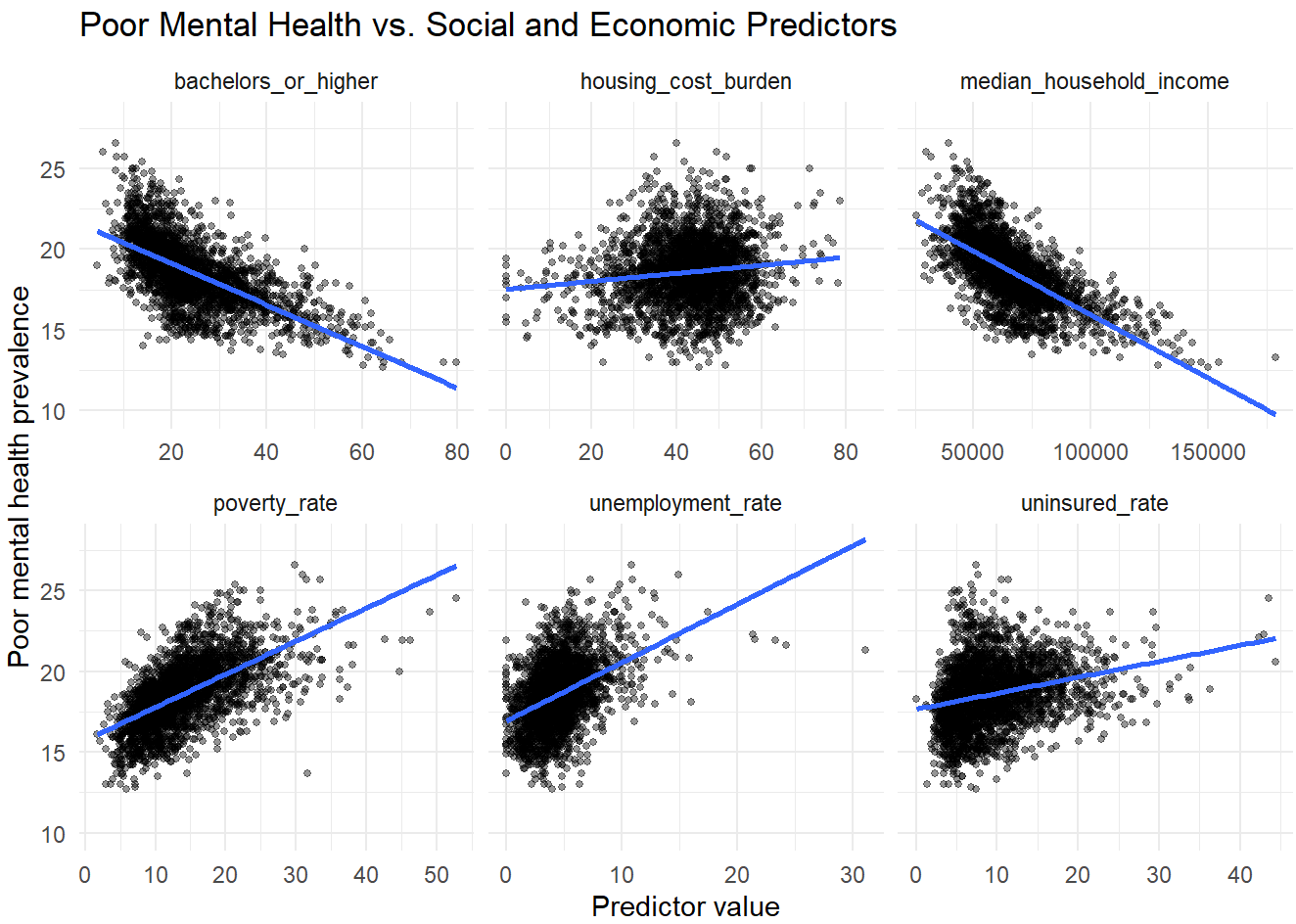
I then create scatterplots of poor mental health against each predictor. This provides an initial look at the direction and apparent strength of the relationship between the outcome and each county-level factor.

```
## Scatterplots of poor mental health vs each predictor
analysis_df %>%
  select(
    poor_mental_health,
    poverty_rate,
    median_household_income,
    unemployment_rate,
    bachelors_or_higher,
    uninsured_rate,
    housing_cost_burden
  ) %>%
  pivot_longer(
    cols = -poor_mental_health,
    names_to = "predictor",
    values_to = "value"
  ) %>%
  ggplot(aes(x = value, y = poor_mental_health)) +
  geom_point(alpha = 0.4, size = 1) +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~ predictor, scales = "free_x") +
  labs(
    title = "Poor Mental Health vs. Social and Economic Predictors",
    x = "Predictor value",
    y = "Poor mental health prevalence"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 3034 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 3034 rows containing missing values or values outside the scale range
(`geom_point()`).
```



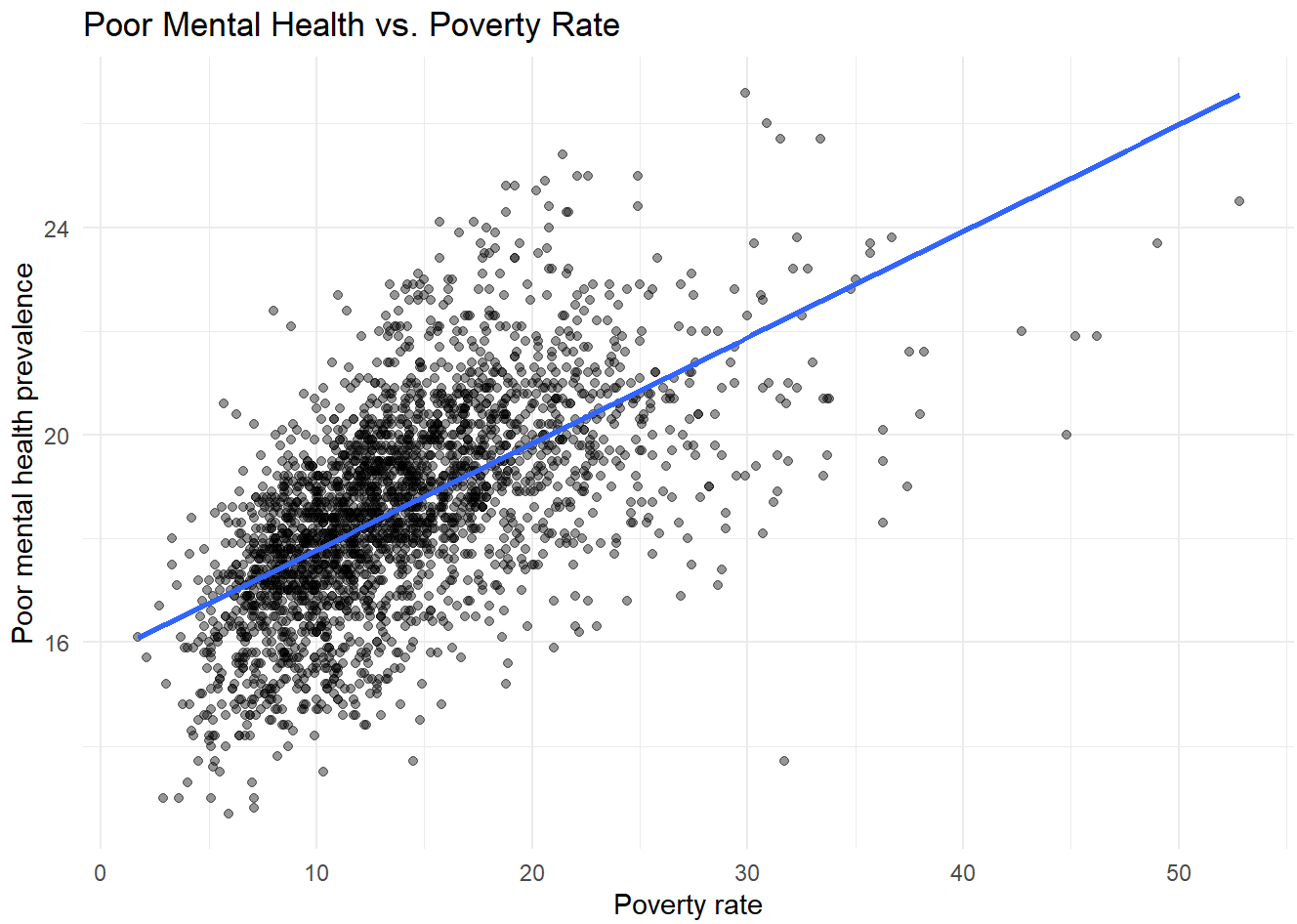
Because some predictors appear more strongly related to poor mental health than others, I also look at a few of the clearest bivariate relationships separately. These focused plots make it easier to inspect the patterns for poverty, income, and education in more detail.

```
# A focused set of scatterplots for the strongest-looking predictors
ggplot(analysis_df, aes(x = poverty_rate, y = poor_mental_health)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Poor Mental Health vs. Poverty Rate",
    x = "Poverty rate",
    y = "Poor mental health prevalence"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 505 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 505 rows containing missing values or values outside the scale range
(`geom_point()`).
```

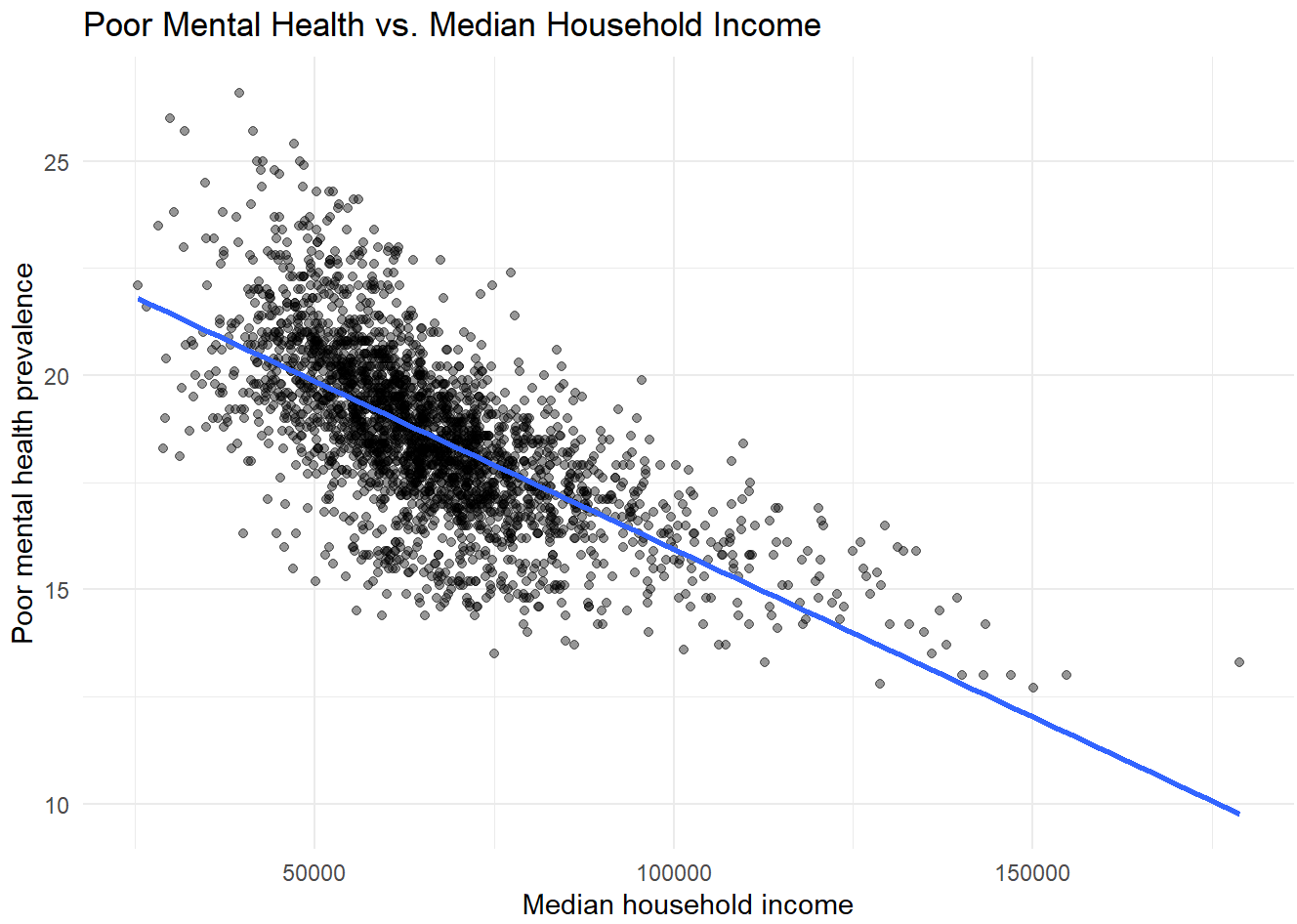



```
ggplot(analysis_df, aes(x = median_household_income, y = poor_mental_health)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Poor Mental Health vs. Median Household Income",
    x = "Median household income",
    y = "Poor mental health prevalence"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 507 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 507 rows containing missing values or values outside the scale range
(`geom_point()`).
```

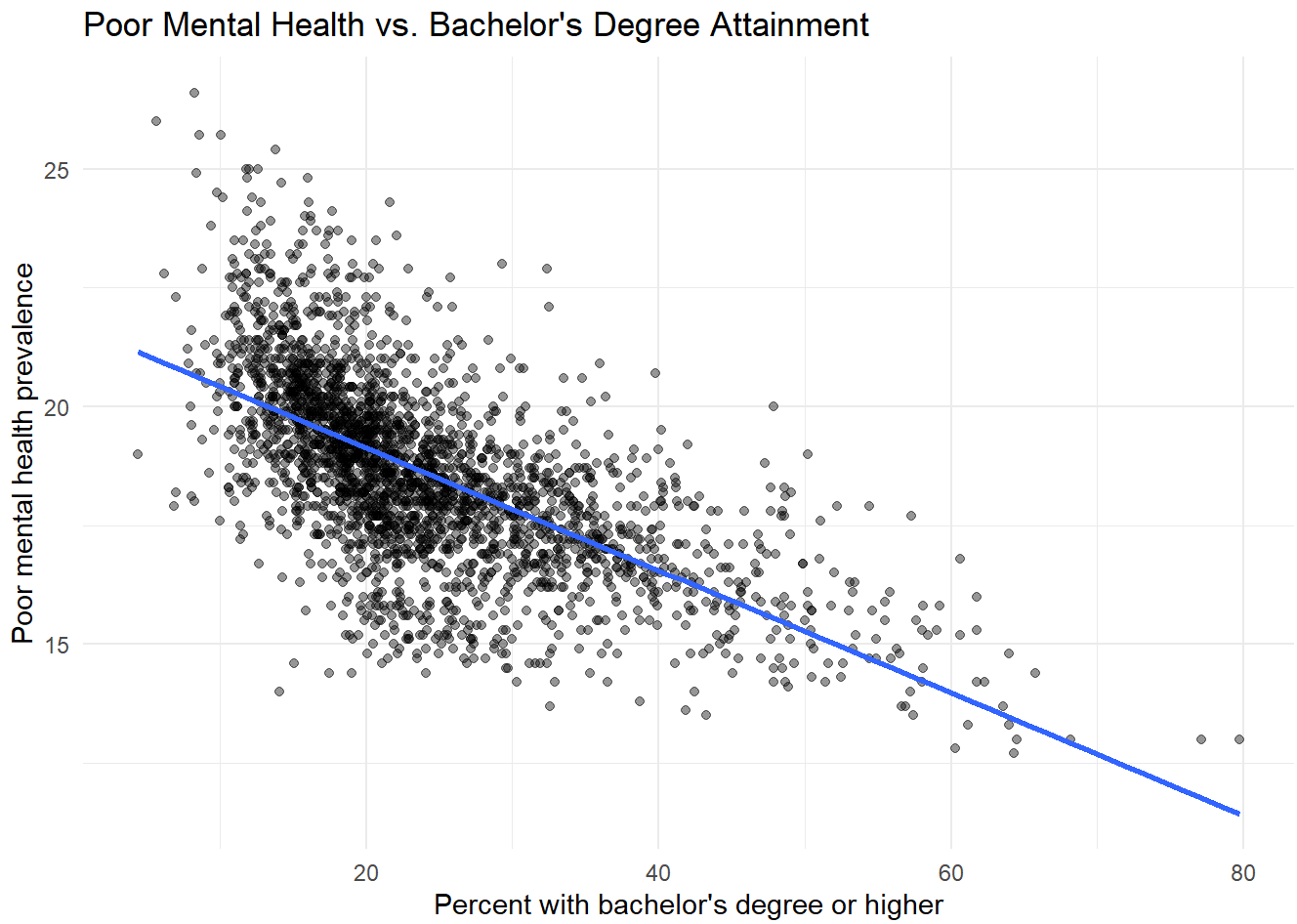



```
ggplot(analysis_df, aes(x = bachelors_or_higher, y = poor_mental_health)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Poor Mental Health vs. Bachelor's Degree Attainment",
    x = "Percent with bachelor's degree or higher",
    y = "Poor mental health prevalence"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 505 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 505 rows containing missing values or values outside the scale range
(`geom_point()`).
```



To understand how the predictors relate to one another as well as to the outcome, I compute a correlation heatmap. This is useful for identifying which variables move together and whether some predictors may capture overlapping dimensions of disadvantage.

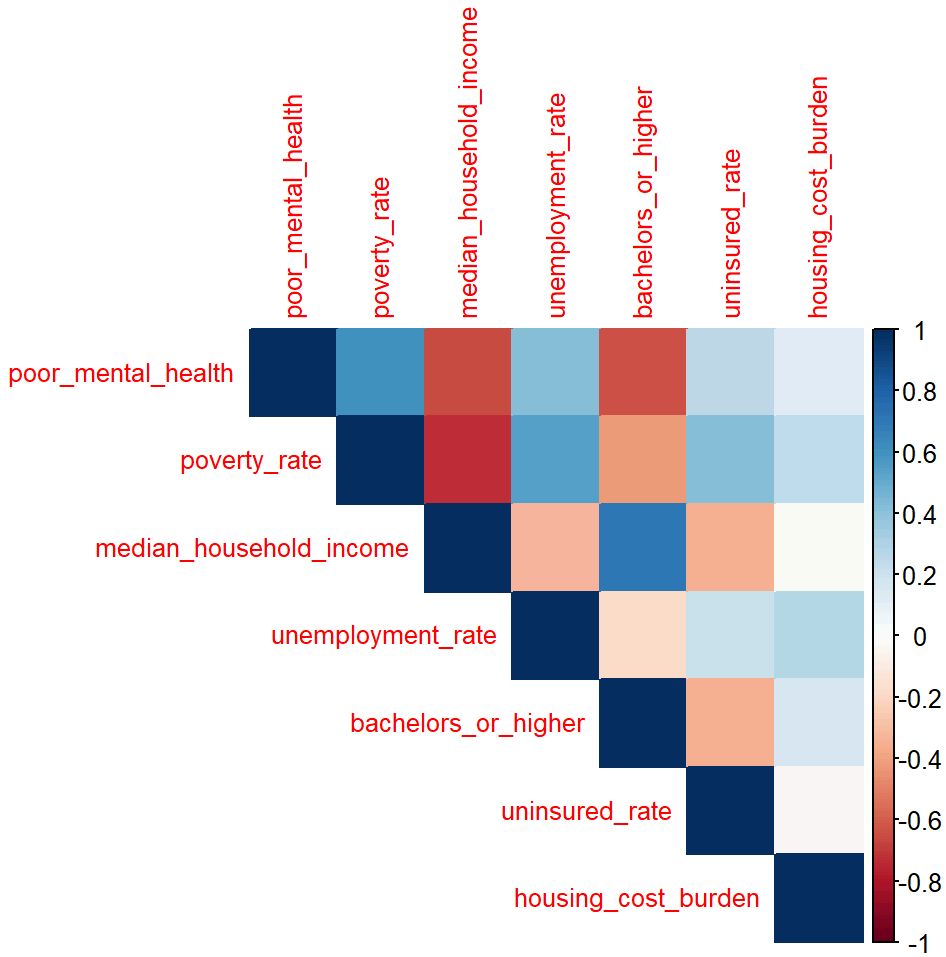
```
# Correlation heatmap
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cor_df <- analysis_df %>%
  select(
    poor_mental_health,
    poverty_rate,
    median_household_income,
    unemployment_rate,
    bachelors_or_higher,
    uninsured_rate,
    housing_cost_burden
  ) %>%
  na.omit()

cor_matrix <- cor(cor_df)
```

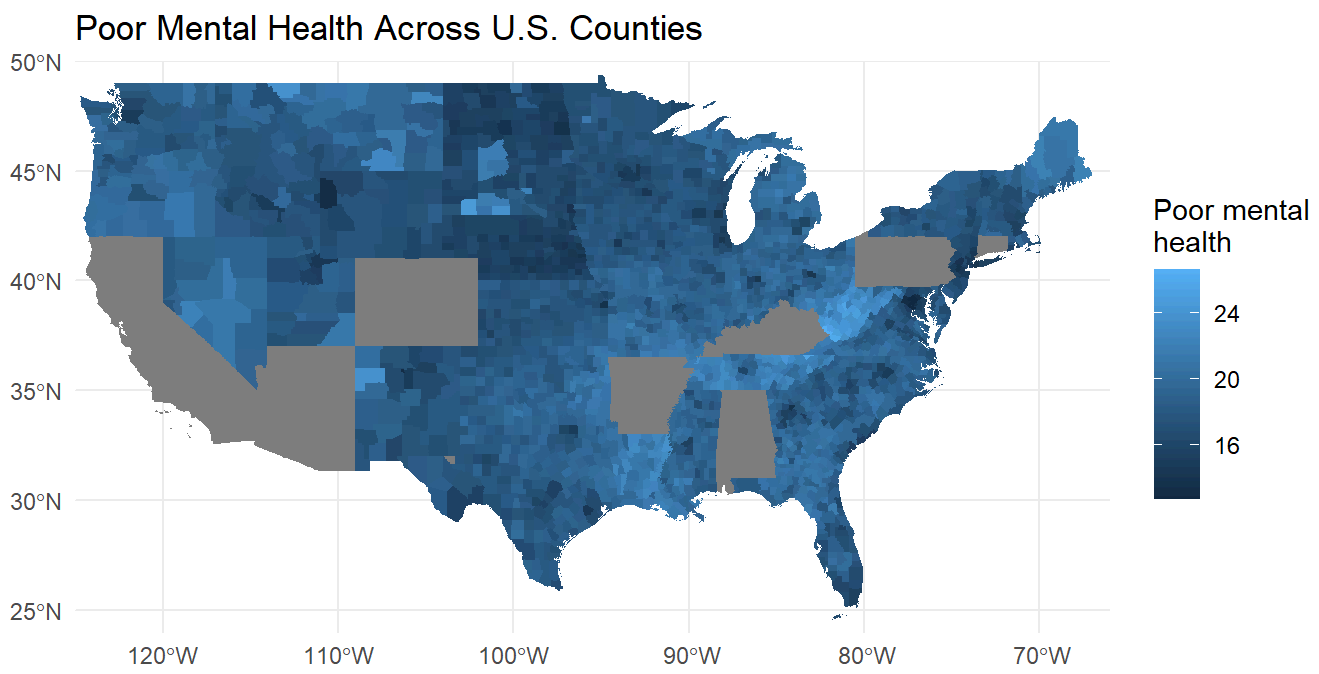
```
corrplot(cor_matrix, method = "color", type = "upper", tl.cex = 0.8)
```



I also map county-level poor mental health to explore whether the outcome shows broad geographic patterning. A choropleth is helpful for seeing whether higher burdens appear clustered in particular parts of the country.

```
# County-level choropleth of poor mental health
library(ggplot2)
library(dplyr)

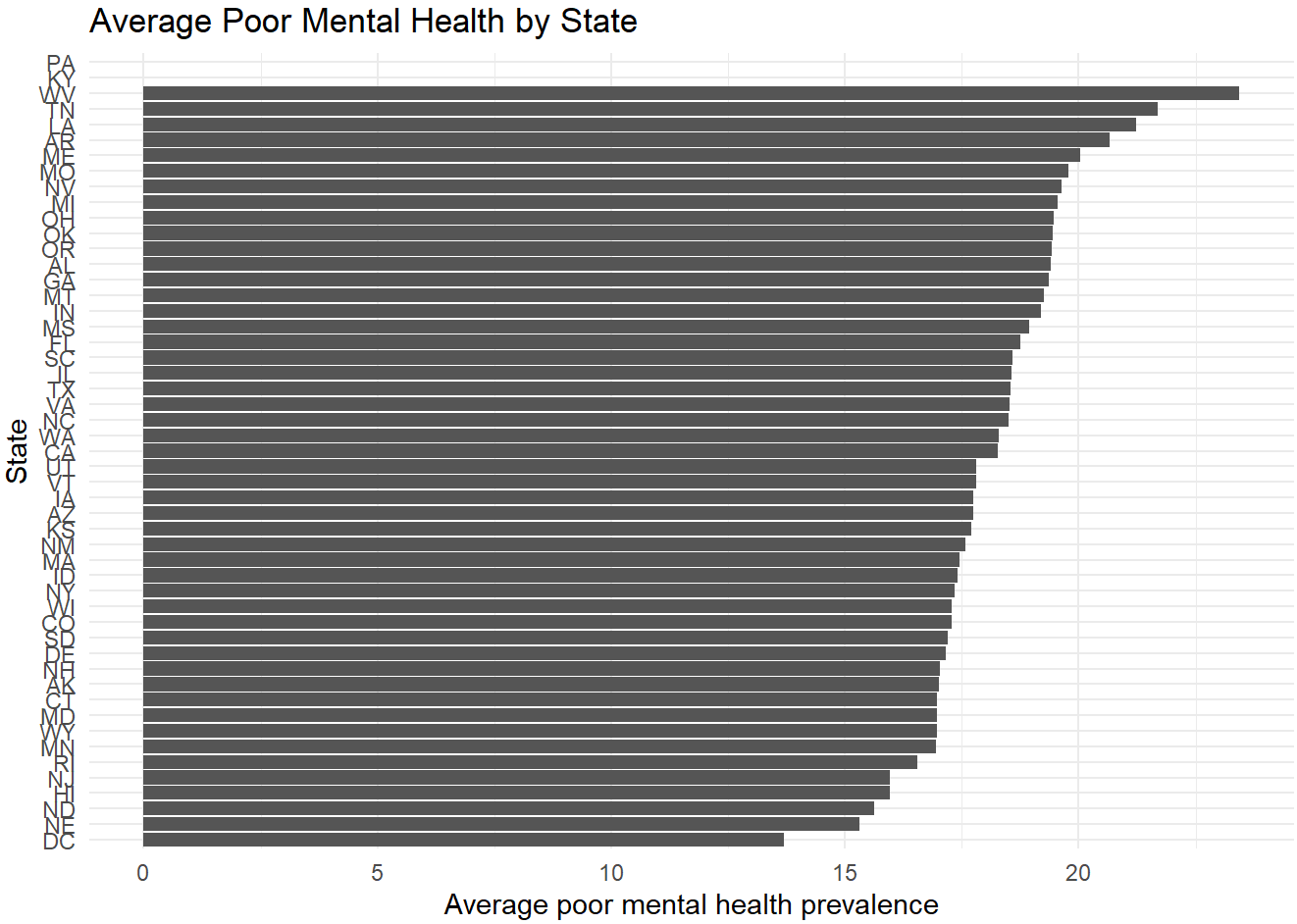
ggplot(map_df) +
  geom_sf(aes(fill = poor_mental_health), color = NA) +
  coord_sf(
    xlim = c(-125, -66),
    ylim = c(24, 50),
    expand = FALSE
  ) +
  labs(
    title = "Poor Mental Health Across U.S. Counties",
    fill = "Poor mental\nhealth"
  ) +
  theme_minimal()
```



Finally, I compare average poor mental health across states as a simple geographic summary. This provides a quick way to see whether some states tend to have systematically higher county-level poor mental health than others.

```
# Quick regional-style comparison using state averages
analysis_df %>%
  group_by(StateAbbr) %>%
  summarize(mean_poor_mental_health = mean(poor_mental_health, na.rm = TRUE)) %>%
  ggplot(aes(x = reorder(StateAbbr, mean_poor_mental_health), y = mean_poor_mental_health)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Average Poor Mental Health by State",
    x = "State",
    y = "Average poor mental health prevalence"
  ) +
  theme_minimal()
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_col()`).
```



Introduction

Mental health is a core part of overall well-being, and it is shaped not only by individual circumstances but also by the broader conditions in which people live. The CDC notes that mental health is influenced by factors at the individual, family, community, and societal levels, which aligns with the social determinants of health framework (Centers for Disease Control and Prevention, “About Mental Health”; Office of Disease Prevention and Health Promotion). Because these conditions affect stress, stability, access to care, and quality of life, they may help explain why some communities experience a higher burden of poor mental health than others.

This project asks: **Which social and economic factors are most strongly associated with poor mental health across U.S. counties?** It also asks a follow-up question: **Do the counties with the worst mental health outcomes simply reflect higher structural disadvantage, or do they appear to have worse mental health than expected even after accounting for those factors?** A county-level approach makes it possible to study broad geographic patterns across differences in poverty, education, insurance coverage, and housing strain.

The outcome variable comes from the CDC’s PLACES dataset, which provides model-based local health estimates across the United States. I use the county-level measure Frequent Mental Distress among adults, defined as the percentage of adults reporting that their mental health was “not good” for 14 or more of the

past 30 days (Centers for Disease Control and Prevention, “Health Status | PLACES”). Because this project compares counties, I use the age-adjusted prevalence of this measure.

To examine possible correlates of poor mental health, I merge PLACES data with county-level indicators from the American Community Survey (ACS) 5-year estimates, including poverty, household income, unemployment, educational attainment, insurance coverage, and housing conditions (United States Census Bureau). Together, these measures provide a county-level dataset for assessing how structural social and economic conditions are associated with poor mental health across the United States.

This project does not aim to make causal claims; instead, it aims to identify which county-level conditions are most closely associated with poor mental health and whether the highest-burden counties appear worse than those conditions alone would predict.

Methods

To answer this research question, I combined county-level data from two public sources. The outcome variable comes from the CDC’s 2025 PLACES county dataset, from which I use the age-adjusted prevalence of frequent mental distress among adults (MHLTH_AdjPrev) as my measure of poor mental health. To examine possible social and economic drivers of this outcome, I merged the PLACES data with county-level estimates from the American Community Survey (ACS) 5-year data obtained through the Census API.

The ACS predictors in this analysis are poverty rate, median household income, unemployment rate, percentage of adults with a bachelor’s degree or higher, uninsured rate, and housing cost burden. These variables capture several major dimensions of community conditions, including economic hardship, labor market stability, educational attainment, access to health insurance, and housing-related financial strain. By joining the PLACES and ACS datasets using county FIPS/GEOID identifiers, I created a county-level dataset to examine how structural social and economic factors are associated with poor mental health across U.S. communities.

Results

County-Level Mental Health Overview

I first present a publication-ready table showing counties with the highest levels of poor mental health. This helps identify which counties appear at the top of the distribution before turning to a national map.

Table 1: Top 15 U.S. Counties by Age-Adjusted Prevalence of Frequent Mental Distress

State	County	Poor Mental Health (%)	Population
West Virginia	Mingo	26.6	22,023
West Virginia	McDowell	26.0	17,439
Tennessee	Hancock	25.7	6,956
West Virginia	Calhoun	25.7	5,959

State	County	Poor Mental Health (%)	Population
West Virginia	Logan	25.4	30,827
West Virginia	Webster	25.0	8,045
West Virginia	Clay	25.0	7,783
West Virginia	Wyoming	25.0	20,277
West Virginia	Lincoln	24.9	19,701
West Virginia	Braxton	24.8	12,162
West Virginia	Summers	24.8	11,581
West Virginia	Roane	24.7	13,743
South Dakota	Oglala Lakota	24.5	13,434
Tennessee	Scott	24.4	22,171
West Virginia	Barbour	24.4	15,378

The table shows that the counties with the highest prevalence of poor mental health are heavily concentrated in West Virginia, with a few additional counties from Tennessee and South Dakota. The percentages are fairly tightly clustered, ranging from about 24.4% to 26.6%, which suggests that the highest-burden counties are all experiencing similarly elevated levels of frequent mental distress.

Next, I map county-level poor mental health across the United States using an interactive choropleth map. This visualization makes it easier to see broad geographic patterns and identify regions where poor mental health appears especially high or low.

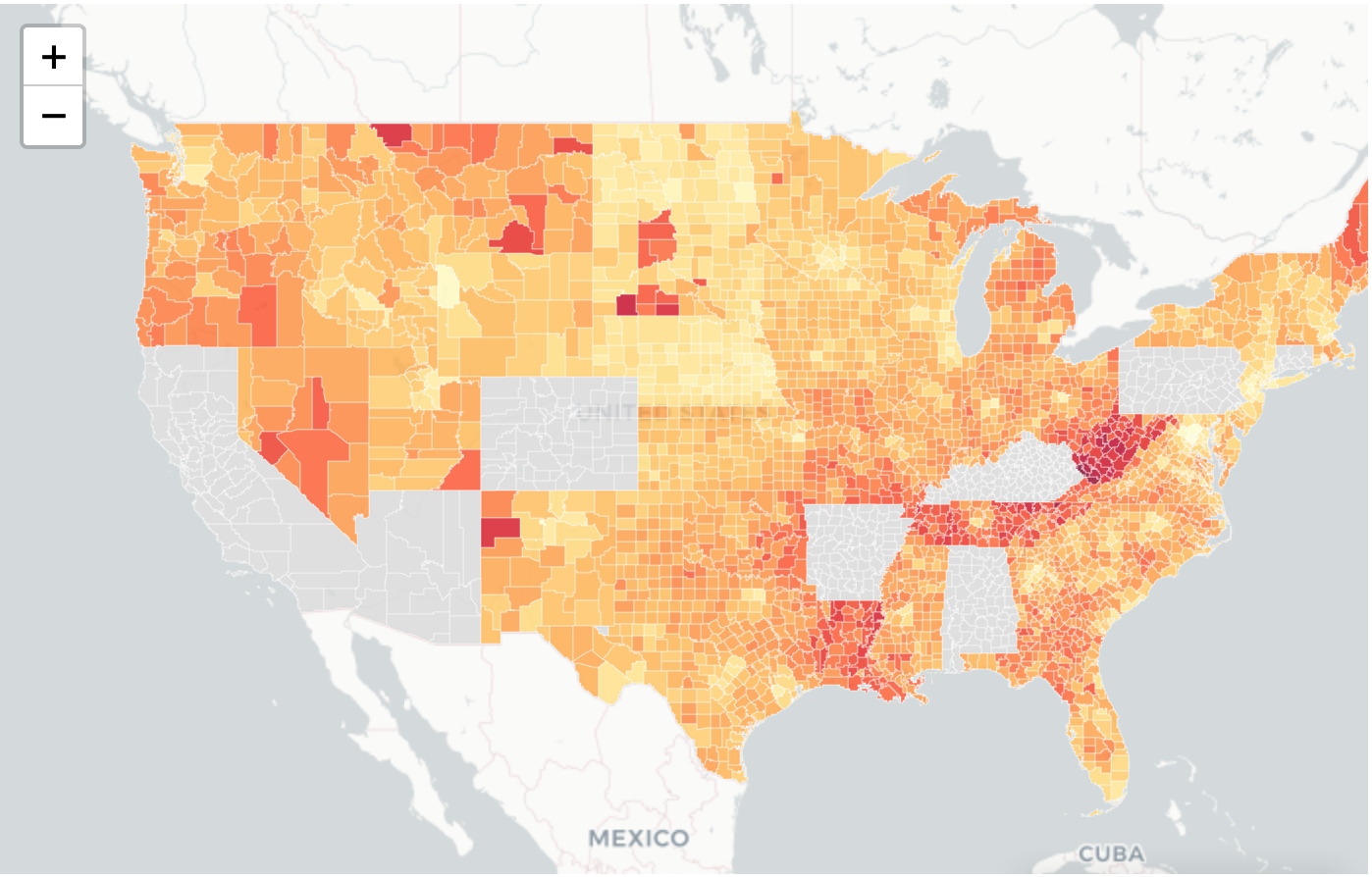


Figure 1

The map suggests broad spatial clustering, with relatively high levels of poor mental health concentrated in parts of Appalachia, the Southeast, and portions of the rural South, while lower levels appear more common in parts of the Great Plains, Upper Midwest, and some Northeastern counties. Overall, the choropleth indicates that poor mental health is not randomly distributed, but instead shows clear regional patterns across U.S. counties.

Associations Between Poor Mental Health and County-Level Predictors

To explore how poor mental health varies with each county-level predictor, I next create a set of side-by-side scatterplots. Faceting the plots allows for direct visual comparison of the direction and strength of the relationship between poor mental health and each predictor.

Figure 2: Scatterplots of poor mental health versus county-level social and economic predictors.

The scatterplots suggest that counties with higher poverty, unemployment, uninsured rates, and housing cost burden tend to have higher levels of poor mental health, while counties with higher median household income and higher educational attainment tend to have lower levels of poor mental health. The clearest patterns appear for poverty rate, median household income, unemployment rate, and bachelor's degree attainment, while the relationship with housing cost burden looks weaker and more diffuse.

Regional Differences in Poor Mental Health

To examine whether poor mental health differs across broad parts of the country, I group counties into U.S. Census regions and compare their distributions using boxplots. This helps show whether some regions tend to have higher or lower county-level burdens of poor mental health, beyond what is visible in the national map.

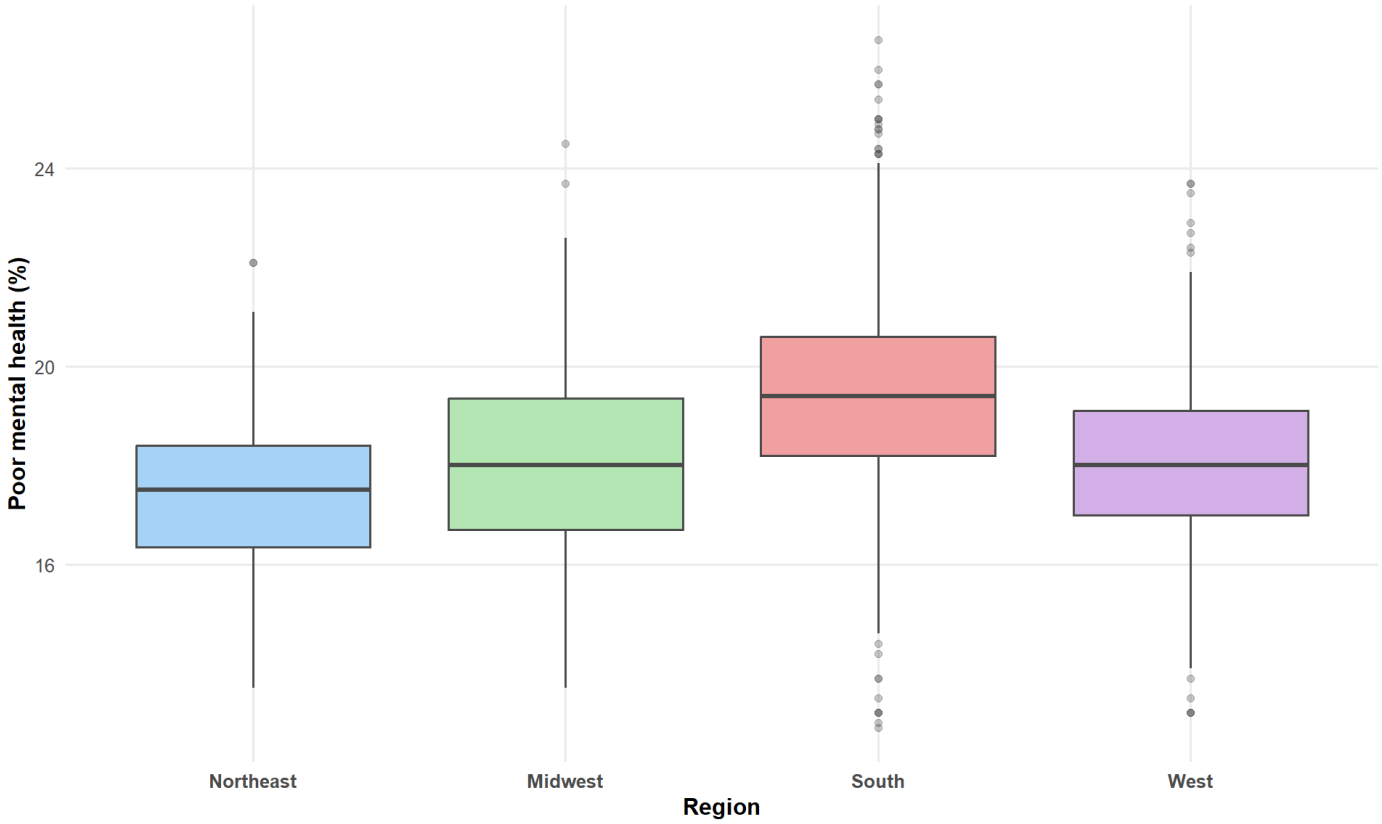


Figure 3: Distribution of poor mental health across U.S. Census regions.

The boxplots show that counties in the South tend to have the highest median levels of poor mental health and also a somewhat wider spread than the other regions. The Northeast appears to have the lowest median, while the Midwest and West fall in between, suggesting meaningful regional differences in county-level mental health burden.

How Do the Highest-Burden Counties Compare to the National Pattern?

To connect the focused analysis of the highest-burden counties back to the full national sample, I compare each county's observed poor mental health to the value predicted by the regression model. Counties above the 45-degree line have worse mental health than expected given their structural disadvantage, while counties below the line have better mental health than expected.

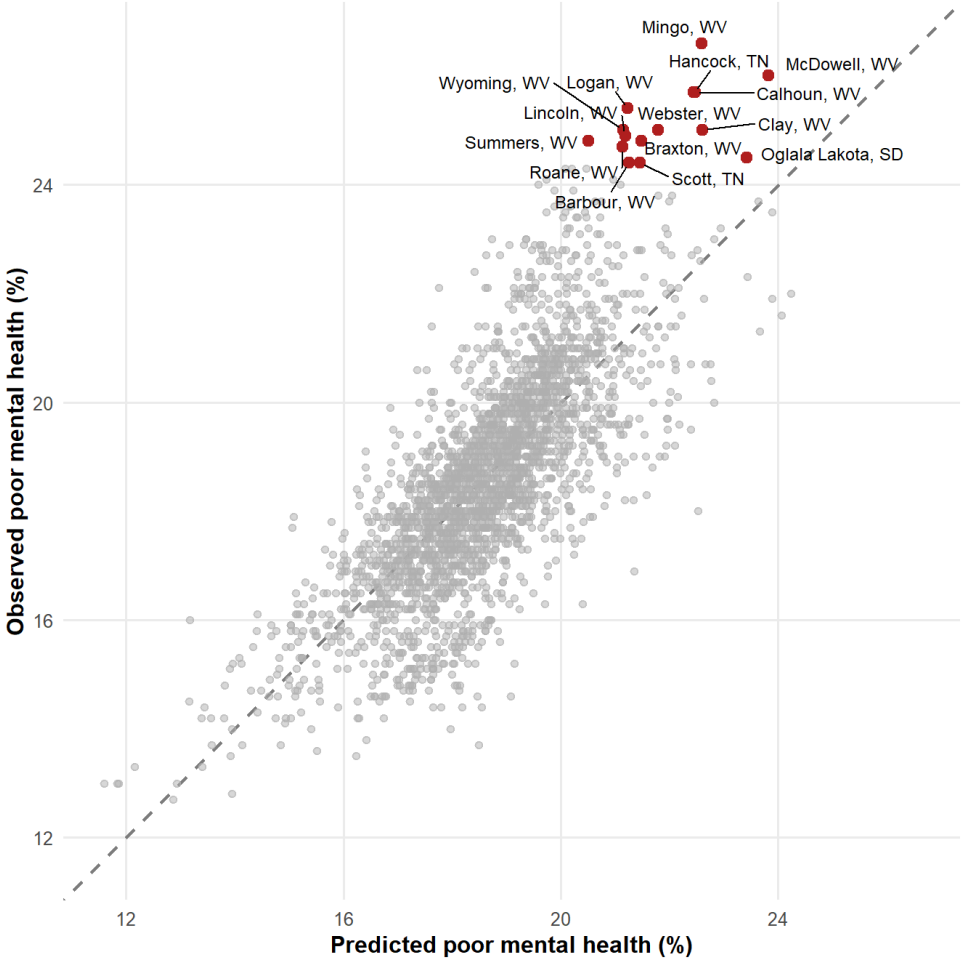


Figure 4: Observed versus predicted poor mental health for all U.S. counties, with the top 15 highest-burden counties highlighted.

The highlighted counties fall mostly above the 45-degree line, indicating that their observed poor mental health is generally higher than what the regression model predicts based on structural disadvantage alone. This suggests that while factors like poverty, income, education, unemployment, insurance, and housing burden explain part of their high mental health burden, many of these counties—especially in West Virginia—still appear unusually unhealthy even relative to similar counties across the United States.

How Do the Most Unhealthy Counties Differ from U.S. Counties Overall?

To compare the most unhealthy counties to the national county distribution, I standardize each predictor across all counties and then calculate the average standardized value for the top 15 counties and for all U.S. counties. I display those averages in a dumbbell plot so the gap between the two groups is easier to compare across predictors on a common scale.

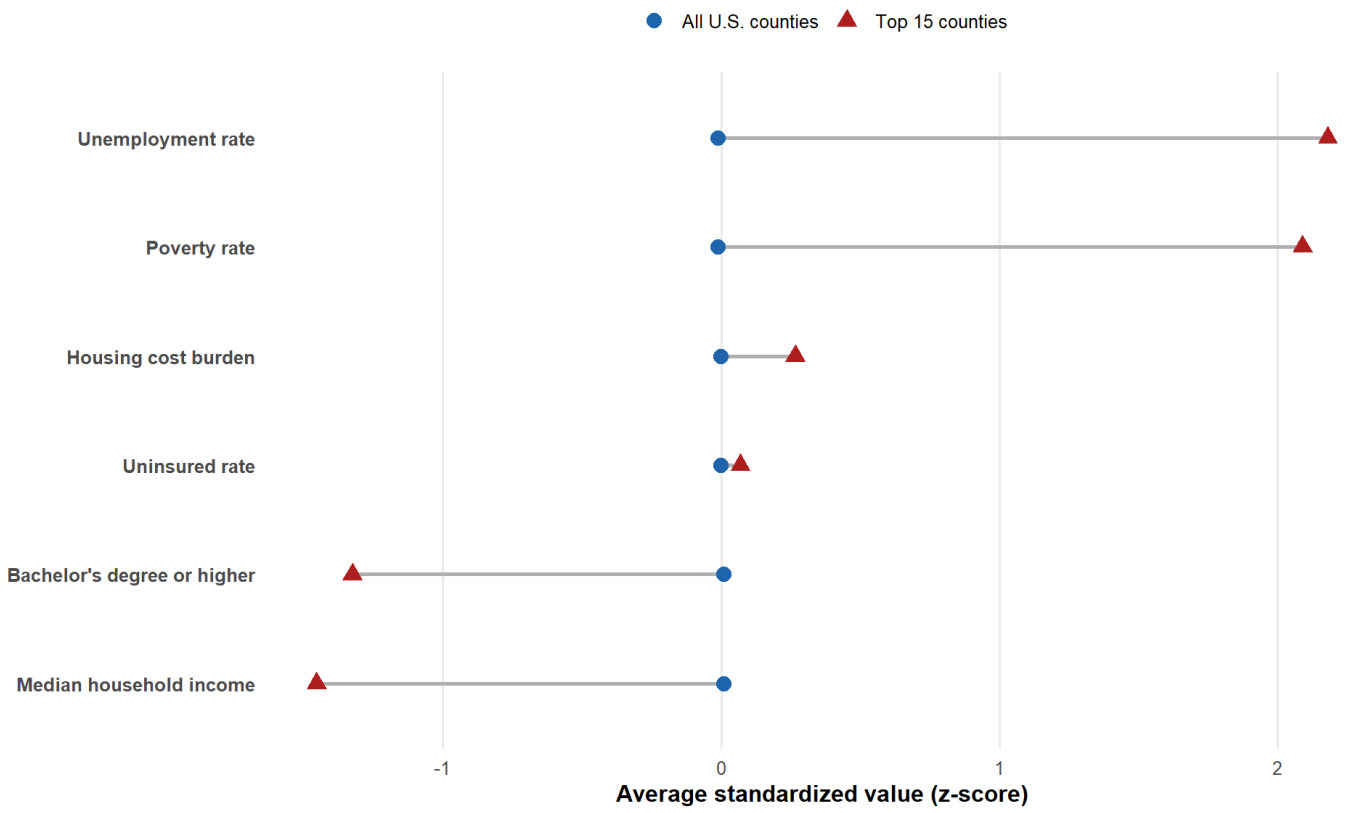


Figure 5: Dumbbell plot comparing standardized predictor averages for the top 15 highest-burden counties and all U.S. counties.

Which Factors Are Most Strongly Associated with Poor Mental Health Across U.S. Counties?

To directly answer the main research question, I fit a multiple linear regression model using all six county-level predictors and compare their standardized coefficients. I display the results as a diverging lollipop plot so it is easier to compare the relative strength and direction of each association on a common scale.

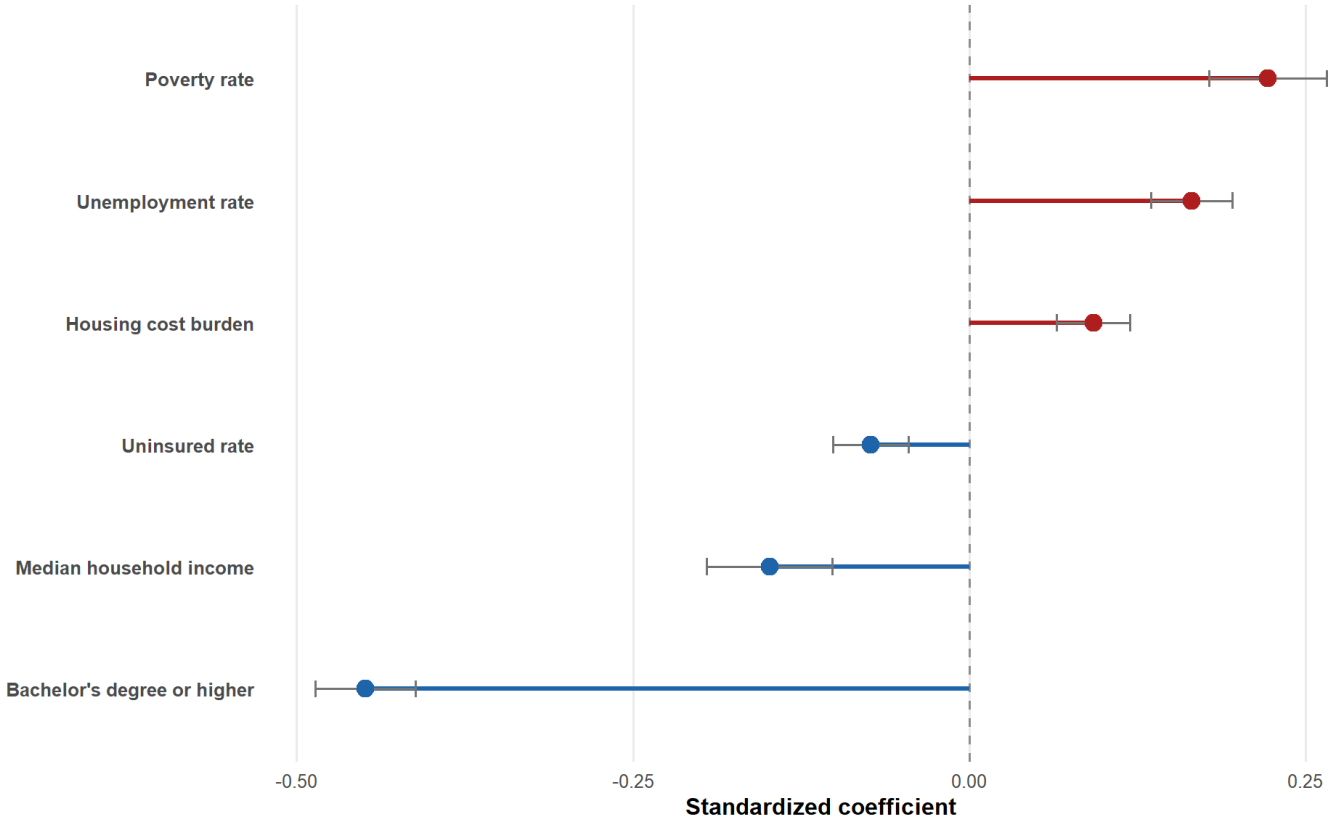


Figure 6: Diverging lollipop plot of standardized coefficients from a multiple linear regression model predicting poor mental health across U.S. counties.

The diverging lollipop plot shows that poverty rate has the strongest positive association with poor mental health, while bachelor's degree attainment has the strongest negative association. Unemployment rate and housing cost burden also have meaningful positive associations, while median household income and uninsured rate are negatively associated, indicating that economic hardship and lower educational attainment are the factors most strongly linked to poor mental health across U.S. counties.

Conclusion and Summary

This project examined which social and economic factors are most strongly associated with poor mental health across U.S. counties. Poor mental health showed clear geographic clustering, with especially high burdens in parts of Appalachia and the South, and many of the highest-burden counties were located in West Virginia.

Across the analysis, counties with higher poverty, unemployment, and housing cost burden tended to have worse mental health outcomes, while counties with higher income and higher educational attainment tended to have better outcomes. The final standardized coefficient plot suggested that poverty rate had the strongest positive association with poor mental health, while bachelor's degree attainment had the strongest negative association. Overall, these results indicate that economic hardship and lower educational attainment are the factors most strongly linked to poor mental health in this county-level analysis.

At the same time, several of the highest-burden counties still appeared to have worse mental health than predicted by these structural factors alone, suggesting that the model does not capture everything driving county-level differences. Future research could examine additional county-level factors, such as access to

mental health care, rural isolation, or social support, to better explain why some counties experience especially high burdens of poor mental health.

Works Cited

Centers for Disease Control and Prevention. "About Mental Health." *CDC*, 9 June 2025, www.cdc.gov/mental-health/about/index.html.

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